Auteur: Nils Van de Velde
Studierichting: Informatica
Oefeningenreeks 5A nr. 7.2

$$\int \sin^9 x dx = \int \sin^8 x d(-\cos x) = \int (1-t^2)^4 dt = -\left(t - \frac{4t^3}{3} + \frac{6t^5}{5} - \frac{4t^7}{7} + \frac{t^9}{9}\right) + c$$

$t = \cos x$

$$\int_0^{\frac{\pi}{2}} \sin^9 dx = 0 - \left(-1 + \frac{4}{3} - \frac{6}{5} + \frac{4}{7} - \frac{1}{9}\right) = \frac{128}{315}$$

References